

Lab 12 | Snowflake CLI

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Snowflake Fundamentals | Day 3 | Estimated time: 20-30 minutes

Lab script: 12-SnowflakeCLI-Lab.sql

You will: create and use a named connection, locate its TOML settings, run a query and a SQL file, and recognize the context of a CLI session.

Before you start: complete the Student Setup Guide, install Snowflake CLI, and run SetUp/SCHEMA-SETUP-SCOTT.sql. Your class role must have access to COMPUTE_WH and DEMO_DB.SCOTT.DEPT. Use instructor-provided alternatives if these names differ.

Download this PDF and the SQL script to your computer. A repository in Snowflake Workspace does not create a local copy. Open the PDF in your browser or PDF reader. Run all commands beginning with **snow** in a local terminal, not the Workspace SQL editor.

1. Check the installation

```
snow --version  
snow --help
```

Check: a version number and command help appear. If snow is not found, open a new terminal after installation and check the setup guide.

2. Create the training connection

```
snow connection list  
snow connection add -n training
```

If training already exists from setup, reuse it and skip the add command. **training is a local connection name**, not a database, warehouse, or Snowflake account.

In the wizard enter your organization-account identifier, username, class role (normally SYSADMIN), warehouse COMPUTE_WH, database DEMO_DB, and schema SCOTT. Leave the password empty; supply it in Step 3. Accept defaults for unrelated optional fields. Use the authentication method required for your account; externalbrowser is only for configured SSO.

Check: training appears in snow connection list. Creating the profile does not prove that login works.

3. Supply credentials and test

For password authentication in Windows PowerShell, replace YOUR_USERNAME and run:

```
$trainingCredential = Get-Credential -UserName 'YOUR_USERNAME'
$env:SNOWFLAKE_CONNECTIONS_TRAINING_PASSWORD = `
    $trainingCredential.GetNetworkCredential().Password
```

Enter your Snowflake password in the credential prompt. The backtick continues the PowerShell command onto the next line. Do not place your password in a command or lab file.

For macOS/Linux, start Bash with **bash**, then run this instead:

```
read -r -s -p "Snowflake password: " SNOWFLAKE_CONNECTIONS_TRAINING_PASSWORD
printf "\n"
export SNOWFLAKE_CONNECTIONS_TRAINING_PASSWORD
```

Configured SSO users skip the password commands. Complete MFA as required by your account; a password alone may not satisfy its authentication policy. If login fails, ask the instructor to confirm the required method.

```
snow connection test -c training
```

Check: the connection test succeeds and identifies your intended account and user. The -c option selects the saved profile. Keep this terminal open for the remaining steps.

4. Locate the saved settings

```
snow --info
```

Locate the configuration file path shown by this command and open it locally in a text editor. A profile in config.toml has this shape:

```
[connections.training]
account = "yourorg-youraccount"
user = "youruser"
role = "SYSADMIN"
warehouse = "COMPUTE_WH"
database = "DEMO_DB"
schema = "SCOTT"
```

These are placeholders, not credentials to use. If connections.toml exists in the same directory, connections come from that file instead; its section is **[training]**. A saved password entry is plain text. This lab omits it. The environment variable also holds a plain-text secret in the current process environment. Do not upload either configuration file or share screenshots containing secrets.

5. Run a query from the terminal

```
snow sql -c training -q "SELECT CURRENT_USER(), CURRENT_ROLE()"
```

Check: one row shows your user and selected role. The -q option supplies SQL directly.

6. Run the supplied SQL script

Change to your local Day3 folder. Replace the example path with the folder where you downloaded the script. Quotes allow spaces in the path.

```
cd "C:\path to your labs\Day3"  
snow sql -c training -f "12-SnowflakeCLI-Lab.sql"
```

On macOS/Linux use your local path, for example **cd "\$HOME/Snowflake Labs/Day3"**, followed by the same snow command. The -f option reads a file on this computer.

Open the script and examine its USE statements and three SELECT statements before running it. They select context, display session information, list departments, and count departments. No objects are created or deleted.

Check: the context query shows DEMO_DB and SCOTT; the department query returns the setup data; DEPARTMENT_COUNT matches the number of departments listed. With the unchanged course setup, expect four departments.

7. Make one small change

Save a local copy as **12-SnowflakeCLI-MyQuery.sql**. In the department-list query, replace **ORDER BY DEPTNO** with **ORDER BY DNAME**. Run your copy:

```
snow sql -c training -f "12-SnowflakeCLI-MyQuery.sql"
```

Check: the same departments are now sorted by name. The department count is unchanged. There is no answer script or submission required.

Session context: each CLI invocation can open a new session. A USE command in one invocation does not establish context for the next. The supplied script sets its context explicitly so that it is repeatable.

8. Finish and review

Remove the password variable when finished, or close the terminal. In PowerShell:

```
Remove-Item Env:SNOWFLAKE_CONNECTIONS_TRAINING_PASSWORD  
Remove-Variable trainingCredential
```

In Bash:

```
unset SNOWFLAKE_CONNECTIONS_TRAINING_PASSWORD
```

Skip these commands if you used SSO and did not create the variables. Keep the training profile for the Internal Stages lab. No database cleanup is needed.

You have completed the lab when you can:

- Explain what training names and what -c, -q, and -f do.
- Locate the active connection configuration.
- Run the original SQL file and your modified copy.
- Explain why a local file is needed even when using Workspace.

Troubleshooting

Connection not found: run `snow connection list` and `snow --info`; check the profile name and active configuration file.

Login failure: confirm account, user, password variable, and the required authentication/MFA method with the instructor.

File not found: check your current local folder and quote the path.

Object or privilege error: confirm schema setup completed in this account and your role can access the database, schema, table, and warehouse.

Network error: follow the firewall/proxy checks in the Student Setup Guide; do not disable certificate validation.

Next lab: 13-InternalStagesLab.sql introduces local file transfers with PUT and GET.

Reference

Official Snowflake documentation:

[Managing connections](#)

[Executing SQL](#)

See `SetUp/Student-Setup-Guide.md` for installation and environment preparation.